

```
39
40 def create_and_insert_bot_frame(bot_response):
41     botFrame = Frame(chatWindow, bg="ffffffd0")
42     Label(
43         botFrame,
44         text=bot_response,
45         font=("Arial", 11),
46         bg="ffffffd0",
47         wraplength=400,
48         justify="left"
49     ).grid(row=0, column=0, sticky="w", padx=5, pady=5)
50     Label(
51         botFrame,
52         text=datetime.now().strftime("%H:%M"),
53         font=("Arial", 7),
54         bg="ffffffd0"
55     ).grid(row=1, column=0, sticky="w")
56
57 chatWindow.insert('end', '\n ', 'tag-left')
58 chatWindow.window_create('end', window=botFrame)
59 chatWindow.insert(END, "\n\n" + "")
60
61 sendButton = Button(
62     baseWindow,
63     font=("Verdana", 12, 'bold'),
64     text="Send",
65     bg="#fd94b4",
66     activebackground="fff467e",
67     fg="ffffff",
68     command=send)
69 sendButton.bind("<Button-1>", send)
70 baseWindow.bind("<Return>", send)
71
72 # Create the main application window using Tk()
73 baseWindow = Tk()
74
75 # Set the title of the window
76 baseWindow.title("The Simple Bot")
77
78 # Set the size of the window
79 baseWindow.geometry("500x250")
```

Guide A Simple Chatbot in Python

Implementing functions for the GUI

At this point, the `sendButton` doesn't do anything. Let's define a function called `send` which should do the following:

- Collect the `user_input` from the `textBox`
- Get the `bot_response`
- Insert both of these into the `chatWindow`

Paste the following code in the top left pane, right under the import statements:

```
def send(event):
    chatWindow.config(state=NORMAL)

    user_input = userEntryBox.get("1.0", 'end-2c')
    user_input_lc = user_input.lower()
    bot_response = get_bot_response(user_input_lc)

    create_and_insert_user_frame(user_input)
    create_and_insert_bot_frame(bot_response)

    chatWindow.config(state=DISABLED)
    userEntryBox.delete("1.0", "end")
    chatWindow.see('end')
```

Once we have the `user_input` and `bot_response`, we'll write a couple of functions to insert them into the `chatWindow`.

Paste the following code in the top left pane, right above the `send` function:

```
def create_and_insert_user_frame(user_input):
    userFrame = Frame(chatWindow, bg="d0ffff")
    Label(
        userFrame,
        text=user_input,
        font=("Arial", 11),
        bg="d0ffff").grid(row=0, column=0, sticky="w", padx=5, pady=5)
    Label(
        userFrame,
        text=datetime.now().strftime("%H:%M"),
        font=("Arial", 7),
        bg="d0ffff"
    ).grid(row=1, column=0, sticky="w")

    chatWindow.insert('end', '\n ', 'tag-right')
    chatWindow.window_create('end', window=userFrame)

def create_and_insert_bot_frame(bot_response):
    botFrame = Frame(chatWindow, bg="ffffd0")
    Label(
        botFrame,
        text=bot_response,
        font=("Arial", 11),
        bg="ffffd0",
        wraplength=400,
        justify="left"
    ).grid(row=0, column=0, sticky="w", padx=5, pady=5)
    Label(
        botFrame,
        text=datetime.now().strftime("%H:%M"),
        font=("Arial", 7),
        bg="ffffd0"
    ).grid(row=1, column=0, sticky="w")

    chatWindow.insert('end', '\n ', 'tag-left')
    chatWindow.window_create('end', window=botFrame)
    chatWindow.insert(END, "\n\n" + "")
```

Now, we want to bind the `send` function to the `sendButton`. We can also bind the `Enter` key to the window so we don't have to click on the button every time.

Replace the `sendButton` initialization in the top left pane with the following code:

```
sendButton = Button(
    baseWindow,
    font=("Verdana", 12, 'bold'),
    text="Send",
    bg="#fd94b4",
    activebackground="fff467e",
    fg="ffffff",
    command=send)
sendButton.bind("<Button-1>", send)
baseWindow.bind("<Return>", send)
```

▼ Code

Your code should look like this:

```
import tkinter scrolledtext as tks #creates a scrollable text window
from datetime import datetime
from tkinter import *

def create_and_insert_user_frame(user_input):
    userFrame = Frame(chatWindow, bg="d0ffff")
    Label(
        userFrame,
        text=user_input,
        font=("Arial", 11),
        bg="d0ffff").grid(row=0, column=0, sticky="w", padx=5, pady=5)
    Label(
        userFrame,
        text=datetime.now().strftime("%H:%M"),
        font=("Arial", 7),
        bg="d0ffff"
    ).grid(row=1, column=0, sticky="w")

    chatWindow.insert('end', '\n ', 'tag-right')
    chatWindow.window_create('end', window=userFrame)

def create_and_insert_bot_frame(bot_response):
    botFrame = Frame(chatWindow, bg="ffffd0")
    Label(
        botFrame,
        text=bot_response,
        font=("Arial", 11),
        bg="ffffd0",
        wraplength=400,
        justify="left"
    ).grid(row=0, column=0, sticky="w", padx=5, pady=5)
    Label(
        botFrame,
        text=datetime.now().strftime("%H:%M"),
        font=("Arial", 7),
        bg="ffffd0"
    ).grid(row=1, column=0, sticky="w")

    chatWindow.insert('end', '\n ', 'tag-left')
    chatWindow.window_create('end', window=botFrame)
    chatWindow.insert(END, "\n\n" + "")

def send(event):
    chatWindow.config(state=NORMAL)

    user_input = userEntryBox.get("1.0", 'end-2c')
    user_input_lc = user_input.lower()
    bot_response = get_bot_response(user_input_lc)

    create_and_insert_user_frame(user_input)
    create_and_insert_bot_frame(bot_response)

    chatWindow.config(state=DISABLED)
    userEntryBox.delete("1.0", "end")
    chatWindow.see('end')

baseWindow = Tk()
baseWindow.title("The Simple Bot")
baseWindow.geometry("500x250")

chatWindow = tks.ScrolledText(baseWindow, font="Arial")
chatWindow.tag_configure('tag-left', justify='left')
chatWindow.tag_configure('tag-right', justify='right')
chatWindow.config(state=DISABLED)

sendButton = Button(
    baseWindow,
    font=("Verdana", 12, 'bold'),
    text="Send",
    bg="#fd94b4",
    activebackground="fff467e",
    fg="ffffff",
    command=send)
sendButton.bind("<Button-1>", send)
baseWindow.bind("<Return>", send)

userEntryBox = Text(baseWindow, bd=1, bg="white", width=38, font="Arial")

chatWindow.place(x=1, y=1, height=200, width=500)
userEntryBox.place(x=3, y=202, height=27)
sendButton.place(x=430, y=200)

baseWindow.mainloop()
```

Which of the following are TRUE about the `sendButton` and the `send()` function?

Hint: There may be more than one correct answer.

- ☐ The `send()` function inserts the `user_input` and the `bot_response` in the `chatWindow`.
- ☐ `sendButton.bind("<Button-1>", send)` binds the `sendButton` to the `send` function.
- ☐ We cannot bind a keyboard key to the `sendButton`.
- ☐ The `send()` function doesn't accept user input.

The `sendButton` can be bound to both, the `send()` function and a keyboard key. It accepts `user_input`, fetches the `bot_response` and inserts them both in the `chatWindow`.

Check all

Now, the only thing left is to define the `bot_response` function!

Next